

Physics requirements of the Phase I Near Detector

May 2024

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1 Introduction: Science Drivers and overarching requirements

DUNE has three primary physics goals:

1. Measure long-baseline neutrino oscillations, including discovery sensitivity to CP violation and the neutrino mass ordering,
2. Measure few-MeV neutrinos, including supernova and solar neutrinos, and
3. Search for physics beyond the Standard Model.

The primary role of the Near Detector (ND) is to predict the neutrino spectrum of the Far Detectors (FD) in the presence of oscillations, and to constrain systematic uncertainties relevant for long-baseline oscillation physics. This physics goal is the design driver of the ND. The **requirements** of the Near Detector are derived from this.

The Near Detector also enables many searches for physics beyond the Standard Model, as well as measurements of neutrino interactions. The BSM searches make use of the high-intensity beamline to produce new particles, which are then detected at the ND. Similarly, precision cross section measurements rely on the very high statistics of the ND. BSM and cross section physics are not generally a design driver, but it is nonetheless important to understand what system capabilities enable these measurements. In this document, we therefore enumerate **goals** that can be pursued at the ND, which may further guide detector design and motivate additional capabilities.

DUNE uses a wideband neutrino beam to measure oscillations as a function of neutrino energy over a broad range from 0.5 – 5 GeV, with a tail that extends to 10s of GeV. The Far Detectors (FD) are liquid argon time projection chambers, which measures neutrino flavor by identifying either a muon or an electron in a charged-current interaction. The muon is characterized by a long, minimum-ionizing track, approximately 5 meters in length per GeV of kinetic energy. Muons in DUNE can be 100s MeV up to several GeV in the oscillation region, and are typically measured by range in the FD, which is large enough to contain muons even when they are > 10 meters long. The momentum resolution is $\sim 5\%$, dominated by stochastic fluctuations in the ionization profile. Electrons are measured by calorimetry in LAr; the radiation length is 14 cm so most energy is contained within 1.5 meters of the interaction vertex. Hadronic energy in the FD is measured by calorimetry in LAr. The FD is able to contain all charged particles in neutrino interactions because of its large size. Fluctuations in the FD hadronic energy measurement are driven by hadronic shower fluctuations to neutral particles, especially neutrons, which often escape detection.

To measure oscillations, DUNE must predict the neutrino spectrum at the FD as a function of the *reconstructed* neutrino energy. The dominant uncertainties are on the normalization and shape of the neutrino flux, neutrino-argon interaction cross sections, and the detector response. The interaction cross section affects not only the number of events observed, but also the observed spectrum, because different interaction topologies have vastly different response in a LArTPC. Because of oscillations and solid angle effects, the fluxes at the ND and FD are different, and the energy dependence of the cross sections is important. The ND also plays an important role in monitoring the stability of the neutrino beam over time.

At an intensity of 1.2 MW and 1.3 seconds between spills (7.5×10^{13} protons per spill), the neutrino interaction rate at the LBNF near site is one per spill per 10 tons of material. This is an unprecedented rate of interactions, which all occur during the narrow beam spill. Interactions in the rock upstream of the detector hall will also contribute. The interaction rate at the FD for beam neutrinos is minuscule, and there will never be confusion due to pile-up. The ND must therefore be capable of separating the activity from each parent neutrino in order to reproduce FD observables.

Table 1 gives the high-level physics requirements which we discuss below:

ND-R0: To predict the observed spectrum at the FD, the strategy of the DUNE Phase I ND is to measure neutrino interactions in the same way as the FD: muons by range, and electrons and hadrons by calorimetry in a LArTPC. This is the same ND strategy successfully used by the MINOS and NOvA experiments. To match the calorimetric response of the FD, a liquid argon detector medium must be employed at the ND. This is the motivation for ND-LAr. To measure muons by range in LAr over the energy range of interest, with a resolution similar to that of the FD, ND-LAr would need to be ~ 20 meters long in the beam direction. This would then require the hall to be much larger, and have no other benefit, as the statistics in ND-LAr are immense and a very long detector is not otherwise needed. The Muon Spectrometer (TMS) is introduced as a cost-saving measure. It is able to measure the muon momentum with a similar resolution, and can be made far more compact due to its higher density. ND-LAr and TMS form a single measurement system.

ND-R1: ND-LAr will measure ~ 50 million ν_μ CC events per year at 1.2 MW intensity. This will

Label	Name	Requirement	Rationale
ND-R0	Predict the observed E_ν spectrum at FD	ND shall perform measurements to enable the prediction of the observed neutrino spectrum at the FD in each flavor	The primary purpose of the ND is to reduce otherwise large systematic uncertainties in the FD prediction which would propagate to the parameter measurements
ND-R1	Constrain flux, cross section, and detector response uncertainties	ND shall perform measurements to constrain systematic uncertainties on the flux, cross section, and detector response	These are the main sources of systematic uncertainty in predicting the observed FD spectrum.
ND-R2	Constrain backgrounds	ND shall perform measurements to constrain key backgrounds in the FD sample	ND measurements are needed to constrain some irreducible backgrounds in the FD sample.
ND-R3	Energy dependence of observables	ND shall perform measurements with different neutrino energy spectra	The energy dependence of the FD prediction, critical to correctly accounting for oscillation effects, can be probed by making measurements in different neutrino fluxes.
ND-R4	Monitor the neutrino beam	ND shall monitor the time variations in the neutrino beam	Unknown and unconstrained changes to the neutrino flux induced by operational beamline variations introduce systematic uncertainties.
ND-R5	Operate at high rates	The ND shall fulfill its requirements accounting for the high rate and pile up from the LBNF neutrino beam in the Near Detector hall.	Pile up and beam-induced backgrounds are an intrinsic aspect of operating at the ND hall.

Table 1: DUNE Near Detector Requirements

be by far the largest sample of neutrino-argon interactions ever collected. The ND-LAr sample can be subdivided into many exclusive final state topologies (0π , $1\pi^\pm$, $1\pi^0$, multi-pion, etc.) and many kinematic bins, all with very high statistics. This makes it possible to constrain specific components of the neutrino-argon cross section model. The observed CC spectrum is predominantly sensitive to the product of the flux and the cross section. To break this degeneracy, the flux can be constrained directly by isolating neutrino-electron elastic scatters, where the neutrino interacts with an atomic electron rather than the nucleus. This process is purely electroweak and can be calculated exactly at tree level. Because the cross section of this process is known, it can be used to measure the flux, and break degeneracies between cross section and flux uncertainties. To constrain detector response effects at the FD, the ND must measure interactions in the same way in a LArTPC.

ND-R2: The oscillation samples at the FD are ν_μ (disappearance) and ν_e (appearance) CC events. The dominant background come from “wrong-sign” events, which are antineutrinos in the FHC (neutrino mode) beam and neutrinos in the RHC (antineutrino mode) beam. The determination of the mass ordering and measurement of CP violation rely on an asymmetry between electron neutrino and antineutrino appearance, which is diluted by the presence of wrong-sign events. The ND must measure this in order to quantify the dilution. This is achieved by magnetization of TMS. Another irreducible background in the appearance channel is intrinsic electron neutrinos in the beam, which come from muon or kaon decays. Smaller backgrounds arise due to neutral currents. To inform the background prediction at the FD as a function of reconstructed energy, these measurements must be made in a LArTPC.

ND-R3: To probe the energy dependence and take data in different neutrino fluxes, the PRISM technique is used. ND-LAr (and its muon spectrometer, TMS) take data off-axis up to 30 m. Off-axis data can also be combined in linear combinations to produce effective fluxes of essentially arbitrary shape. A particular combination can reproduce an *oscillated* FD flux for a particular oscillation hypothesis, making it possible to perform a data-to-data analysis without extrapolating in neutrino energy.

ND-R4: In order to monitor the neutrino beam stability, a regular cycle of on-axis measurements is required. This is needed because beam focusing changes typically do not impact the far-off-axis flux. The preferred solution is to have a permanent on-axis system (SAND) continuously monitoring the neutrino beam. The minimum requirement is to collect sufficient on-axis statistics to ensure beam stability once per month. This could be achieved by moving ND-LAr+TMS on axis for several days once per month.

ND-R5: The LBNF beam is expected to deliver spills that are 10 μ s in duration. Within one spill, there are 588 “buckets” with a width of several ns and peaks separated by 18.83 ns. The number of protons per spill is expected to be $\sim 7.5 \times 10^{13}$, resulting in an interaction rate of roughly one per spill per 10 tons of material. The ND must be able to resolve individual neutrino interactions, either by isolating particles spatially, or by using timing to separate neutrinos from different buckets. The number of protons per spill is not expected to vary significantly as a function of the average beam intensity. This is because ACE-MIRT achieves an increased intensity by reducing the Main Injector cycle time, resulting in more spills of similar intensity. Therefore, the rate environment is not expected to be substantially different as the intensity ramps up to 2 MW and beyond.

1.1 Goals

In this document, the ND physics requirements flow down from the long-baseline neutrino oscillation measurement goals. These are central to the DUNE physics program, and required in order for DUNE to be successful. Goals are not strictly required for DUNE success, but are each well-motivated, highly impactful physics searches. Goals may not necessarily drive key elements of the

ND design; a detector component or capability that is necessary only to achieve goals could be considered a scope contingency. Goals can be generally divided into high-level categories given in Table 2.

ND-G1	Search for new physics produced in the LBNF beam
ND-G2	Measure neutrino interaction cross sections
ND-G3	Improve the DUNE oscillation program, beyond what is required
ND-G4	Simplify operations or provide operational robustness

Table 2: The main categories of physics goals at the Near Detector.

Many of these measurement goals stem from deploying a low-density tracking detector with passive nuclear targets inside the KLOE magnet and calorimeter as part of SAND. While this is not strictly required for SAND’s role as a beam monitor, it greatly expands the reach of the DUNE physics program. It also creates synergies with ND-LAr+TMS in the core neutrino oscillation measurements by enabling complementary alternative analysis strategies, potentially building in valuable redundancy.

ND-G1: Searching for new physics signals at the Near Detector is a core component of DUNE’s third primary physics goal of searching for physics beyond the Standard Model. This is listed here as a goal, rather than a requirement, because there are no specific BSM searches which are expected to drive the ND design. At the Near Detector, it is expected that these searches will be for particles originating in the beamline; the ND is not expected to be competitive in any search of cosmogenic origin due to its reduced size and overburden. The very high intensity of the LBNF beam creates excellent goals for world-leading rare process searches.

ND-G2: Measurements of neutrino interactions on argon at the Near Detector are critical direct inputs to the DUNE long-baseline oscillation program; oscillation parameters are extracted by jointly fitting Near and Far Detector data. These measurements are required by ND-R0 and ND-R1, and enumerated in Table 3. The ND can also extract differential cross section measurements. For example, DUNE could extract a cross section for $CC1\pi^+$ events from ND-LAr, while also fitting that sample directly in an oscillation fit. Cross sections may also be extracted from data that does not enter into the oscillation fit, but is of interest for informing neutrino interaction modeling, or nuclear physics. For example, SAND could measure a cross section on various nuclei, which would inform the A-dependence of neutrino interactions.

ND-G3: The minimum Near Detector for the Phase I DUNE long-baseline oscillation physics measurements is ND-LAr (and TMS) measuring neutrino-argon interactions both on- and off-axis, and SAND monitoring the neutrino beam stability. There are Near Detector measurements beyond this which could provide valuable additional information. This could enable alternative analysis strategies, or add redundancy to the main analysis. There is significant overlap between ND-O1 and ND-O2; for example, a neutrino-hydrogen cross section would be interesting as a comparison sample for neutrino-nucleus measurements from the standpoint of understanding nuclear structure, but it could also provide valuable complementary constraints on free nucleon cross sections for the long-baseline physics analyses beyond what is required.

ND-G4: This goal does not impact the physics capability. Detector capabilities may be motivated by a desire to simplify the operations of the detector, to increase flexibility in operations, or to add robustness to downtime.

2 Measurement requirements

Measurement requirements for the Phase I ND are given in Table 3.

ND-M0: The FDs are LArTPCs that select ν_μ events by the presence of a long muon track, and ν_e events by the presence of an electromagnetic shower at the vertex. It measures the muon energy by range, and the electron and hadronic energies by calorimetry in liquid argon. The measured spectrum is sensitive to the cross section and detector response models in a convoluted way, such that the only way to constrain this at the Near Detector is to use a LArTPC. TMS fulfills the critical task of muon spectrometry, while requiring less physical space than a single LArTPC detector would need in order to do this. TMS must have momentum resolution comparable to what can be achieved by range in LAr. In order to match the FD, in which there is no beam pile-up, ND-LAr+TMS must also be able to isolate individual neutrino interactions.

ND-M1: Measurements of charged-current interactions at the ND fundamentally constrain a product of the flux and the cross section. This leads to anticorrelations in the flux and cross section uncertainties, which affect the FD prediction differently. To directly measure only the flux, and break this degeneracy, it is necessary to make a direct measurement of a “standard candle” process with known cross section. The only such process in the relevant energy range is $\nu + e$ elastic scattering. The cross section of this process is very small, with an expected event rate of 100 events per ton of fiducial mass per year at 1.2 MW intensity.

ND-M2,3: These requirements pertain to predicting the background in the FD. The two irreducible backgrounds in the FD appearance samples (selected $\nu_e + \bar{\nu}_e$ events) are wrong sign interactions (i.e. $\bar{\nu}_e$ in neutrino mode and ν_e in antineutrino mode) and intrinsic ν_e in the beam. To constrain the wrong-sign oscillated ν_e in the FD, it is necessary to measure wrong-sign ν_μ interactions in the ND. This is important because DUNE must measure the asymmetry in the $\nu_\mu \rightarrow \nu_e$ and $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ oscillation driven by the matter effect and CP violation. With the sign selection capability of TMS, this can be measured in ND-LAr+TMS in the same way as the FD, on the same target. For the intrinsic ν_e background, it is not nearly as important for it to be sign-selected, because the FD measures the sum of $\nu_e + \bar{\nu}_e$. This is because the beam ν_e has only a tiny oscillation effect (due to the smallness of θ_{13}). The intrinsic ν_e can be measured in ND-LAr in the same way as in the FD.

ND-M4: The primary means by which the spectrum at the ND is varied (ND-R3) is with DUNE-PRISM, which exploits the steady decrease and narrowing of neutrino energies as one samples the beam further from the axis. Measuring different spectra across the energy range of interest for neutrino oscillation measurements (roughly between the first and second maxima) is needed to constrain the energy dependence in the interaction model, and enables a robust prediction of the FD spectra in the presence of oscillations.

ND-M5: A beam monitoring system must continuously measure the position and intensity of the proton beam, the positions and current of the focusing horns, and the flux of muons from the pion and kaon decays that produce the neutrino beam. This system must be capable of detecting a 1% change in the absolute muon rate in real time.

ND-M6: Subtle changes in the neutrino flux are not expected to be detectable using the muon monitoring system. Many such effects would also not be detectable off-axis, because the off-axis flux is less affected by the focusing apparatus. The ND must therefore make an on-axis measurement of the neutrino beam position, rate, and spectrum each month, to ensure that it has not changed and the FD data taken that month can be analyzed. If a beam change is detected in real time by the muon monitors, it must be possible to bring the ND on axis within ~ 1 day to perform a measurement of the neutrino beam spectrum for ~ 1 week. Without a permanent on-axis near detector, it is possible that beam changes that are not detectable by the muon flux could occur,

Label	Name	Requirement	Rationale	System	Ref. Goal
ND-M0	Match FD performance	The ND shall classify interactions and measure outgoing particles in a LArTPC with performance matching the FD	To predict the observed FD spectrum, the ND must employ a LArTPC, and measure events in the same way as the FD, with the same resolutions	ND-LAr+TMS	ND-R0, ND-R1
ND-M1	Constrain neutrino flux	ND shall measure $\nu + e$ elastic scattering to constrain flux normalization to $< 2\%$	$\nu + e$ elastic scattering is a “standard candle” process, which can be used to constrain flux and break degeneracies between flux and cross section effects	ND-LAr	ND-R1
ND-M2	Measure wrong-sign contamination	ND shall constrain the rate of wrong-sign interactions to $< 20\%$ in FHC and $< 5\%$ in RHC	Wrong-sign events dilute oscillation asymmetries at the FD, and must be constrained by the ND by measuring the charge sign of outgoing muons	ND-LAr+TMS	ND-R2
ND-M3	Measure the intrinsic ν_e	ND shall constrain the rate of intrinsic ν_e to $< 2\%$	Intrinsic ν_e and $\bar{\nu}_e$ are an irreducible background to ν_e appearance, and must be constrained by measuring ν_e CC in a LArTPC at the ND	ND-LAr	ND-R2
ND-M4	Off-axis measurements	ND shall make measurements in different neutrino fluxes, covering energies from 0.5-3.0 GeV	To probe the energy dependence of neutrino interactions, and predict oscillated FD spectra, ND must move off-axis	ND-LAr+TMS, PRISM	ND-R3
ND-M5	Monitor beam in real time	Muon monitors shall be capable of detecting a 1% change in the integrated muon rate	Muon detectors must detect changes in the flux in real time, which can then be further studied with ND	MuMS	ND-R4
ND-M6	Monitor neutrino beam	ND shall measure on-axis neutrino spectrum at least monthly	The ND must make on-axis measurements regularly and if a change is detected by MuMS; continuous monitoring is preferred	ND-LAr+TMS or SAND	ND-R4
ND-M7	Isolate individual interactions	ND shall identify and segment activity from individual neutrino interactions.	To match FD performance, ND must isolate activity from individual neutrino interactions, despite operating in a high-rate environment	ND-LAr+TMS, SAND	ND-R5

Table 3: Measurement requirements for the ND.

Label	Name	Requirement	Rationale	System	Ref. Goal
ND-X1	Search for Neutral Heavy Leptons (HNL)	ND should search for neutral particles produced in the beamline that decay in the ND	Neutral particles produced in the beamline that decay in the ND can be detected, especially with a low-density detector which could achieve zero background.	SAND, ND-LAr	ND-G1
ND-X2	Search for new physics with $\nu + e$	ND should perform electroweak tests and new physics searches using neutrino-electron elastic scattering	Measuring $\nu + e \rightarrow \nu + e$ at different neutrino energies is sensitive to $\sin^2 \theta_W$, and the neutrino dipole moment. A contribution from light dark matter would enhance the signal at large angle.	ND-LAr, PRISM	ND-G1
ND-X3	Measure neutrino tridents	ND should search for tridents characterized by dilepton final states	Tridents are a pure electroweak SM process that could have an enhanced cross section due to new physics.	ND-LAr, SAND	ND-G1
ND-X4	Search for short baseline oscillations	ND should search for neutrino oscillations with $\Delta m^2 \sim 1 - 100 \text{ eV}^2$	With wideband beam, and relatively well-constrained flux, DUNE has excellent sensitivity to short baseline oscillations, especially at Δm^2 above what will be measured in SBN	ND-LAr, SAND	ND-G1

Table 4: DUNE Near Detector BSM Goals

and it would not be known for up to one month while the ND is off axis. The FD data taken during this period may not be analyzable; for this reason it is preferable to have a permanent on-axis ND component. The relation between ND-M5 and ND-M6 is further discussed in Section 4.3.

ND-M7: Due to the intensity of the neutrino beam, the ND operates in an environment with a high level of external background activity and pileup of neutrino interactions. In order to make the required physics measurements, the ND must be able to reject these backgrounds which are not present at the FD and segment reconstructed objects into individual neutrino interactions, as they would be observed at the FD.

3 Measurement goals

In addition to the requirements for oscillation physics enumerated in Table 3, the ND can contribute to a broader, more robust physics program with additional measurement goals. Table 4 lists the goals pertaining to BSM physics, and Table 5 lists other physics goals.

ND-X1: One of the most interesting potential BSM signatures in DUNE are Heavy Neutral Leptons (HNL), which are produced in the beamline and decay in the ND. The signal rate scales with the detector volume, while the background from ordinary neutrino interactions scales with

Label	Name	Requirement	Rationale	System	Ref. Goal
ND-X5	Measure free proton cross sections	ND should measure $\nu - p$ cross sections	$\nu - p$ measurements, free of nuclear effects, constrain nucleon-level cross section predictions, and can be achieved via the “solid hydrogen” concept	SAND	ND-G2, ND-G3
ND-X6	Measure $\nu - \text{Ar}/\nu - \text{H}$ σ ratios	ND should measure the ratio of ν cross sections on Ar and H(p).	Measuring cross section ratios to free nucleon is directly sensitive to nuclear effects, and could guide theory to improved nuclear models	SAND	ND-G2
ND-X7	Measure cross sections on various targets	ND should measure neutrino cross sections on various nuclear targets with the same detector	Studying the A -dependence of various inclusive and exclusive neutrino cross sections can inform nuclear models.	SAND	ND-G2
ND-X8	Measure $\nu_\mu, \bar{\nu}_\mu, \nu_e, \bar{\nu}_e$ interactions	ND should separately measure interactions of each neutrino flavor in the neutrino beam	SAND can complement the wrong-sign and intrinsic ν_e measurements of ND-LAr with additional information from sign-selecting the $e^\pm$ samples	SAND	ND-G2
ND-X9	Inverse μ decay measurement	ND should identify and measure the rate of inverse μ decay events	Similar to ND-M1, the high-energy tail can be constrained with inverse muon decay. However the μ in process are too energetic to analyze in TMS.	SAND	ND-G3
ND-X10	Continuous on-axis beam monitoring	ND should have continuous on-axis neutrino monitoring	With a fixed on-axis detector, ND-LAr+TMS does not need to return to the on-axis position monthly, and can take extended off-axis data, reducing the required movement rate and reducing the possibility of overlooking a variation in the beam conditions	SAND	ND-G4

Table 5: DUNE ND goals pertaining to neutrino cross sections, enhancements to the long-baseline program, and operational robustness.

mass; thus, a low-density detector is optimal. It has been shown that a background-free search in DUNE ND could be world-leading in sensitivity, reaching the parameter space favored by Type-I see-saw models. A search in ND-LAr (with TMS providing muon spectrometry, as usual) is possible, but unlikely to reach zero background. With a low density inner tracker, SAND could improve the rejection of backgrounds by identifying nuclear breakup, and reduce the background rate by more than an order of magnitude by density alone.

ND-X2: While $\nu + e$ elastic scattering is primarily used as a direct flux constraint (ND-M1), it can also be used for precision electroweak measurements and searches for new physics. The neutrino energy dependence of $\nu + e$ is sensitive to $\sin^2 \theta_W$, and in principle to the dipole moment of the neutrino. Measuring $\nu + e$ in different fluxes with PRISM is a unique way to access this physics. The dependence of the $\nu + e$ rate and spectrum on the off-axis angle is also sensitive to an additional contribution from dark matter-electron scattering. Light dark matter (LDM) produced in the beamline would give a different angle dependence because the dispersion of the LDM and the neutrino beam is different, and the scattering kinematics are different when the probe is massive. Because ND-LAr already has a requirement to measure $\nu + e$ (ND-M1) and move off-axis (ND-M4) for other purposes, this goal does not impose any additional detector requirements.

ND-X3: Neutrino tridents occur when a neutrino scatters off the Coulomb field of the nucleus, generating a pair of charged leptons, for example $\nu_\mu \rightarrow \nu_\mu \mu^+ \mu^-$ or $\nu_\mu \rightarrow \nu_e e^+ \mu^-$. This is a Standard Model process with a tiny cross section, but could be enhanced by a contribution mediated by a Z' . This could be done in ND-LAr (large mass, but no electron charge ID, and worse muon charge ID at high momentum), or in SAND with a capable low density inner tracker (smaller mass, but exquisite reconstruction of charge and momentum).

ND-X4: The event rate at the DUNE ND makes it a compelling place to search for short-baseline oscillations. One particular feature of DUNE is its ability to constrain the flux prediction *in situ* with $\nu + e$ elastic scattering, which is required for the long-baseline physics in ND-M1. As a result of this, the LBNF flux at the ND will be much better constrained than any previous accelerator-based neutrino experiment. For Δm^2 in the few eV^2 region, the two-detector SBN program will reach very small systematic uncertainties using its Near Detector. However at higher Δm^2 , where oscillations are rapid in the ND, DUNE's lower flux uncertainty gives an advantage.

ND-X5-7: SAND presents many opportunities to make interesting cross section measurements. With a capable, low-density inner tracker with CH_2 and C targets, and other thin passive nuclear targets, SAND can pursue a measurement program similar to what has been achieved by MINERvA, but with much higher statistics, much smaller wrong-target backgrounds, and much better resolutions. By including a carbon passive target, SAND can make measurements on "solid hydrogen" by subtracting carbon from the hydrocarbon. Measurements of neutrino-hydrogen cross sections on bubble chambers in the 1970s with a few hundred events continue to be extremely useful for the community because of the uniqueness of having zero nuclear effects. SAND could drastically improve this, and additionally measure cross section ratios on a variety of targets compared to free protons. These measurements would inform our understanding of the A -dependence of neutrino interactions. The hydrogen measurement in particular could also improve the oscillation sensitivity, if better nuclear models can be developed based on this data.

ND-X8,9: With a capable inner tracker, SAND presents many opportunities to complement or potentially improve the oscillation measurements. SAND cannot replace the core functionality of ND-LAr, which is to make measurements of the same quantity (reconstructed neutrino energy in a LArTPC) that is observed in the FD. However, SAND has the additional capability of sign-selecting the electron neutrinos, which could improve the constraint of the intrinsic background. SAND also has sign selection and momentum measurement capability of very high energy muons, which opens the door to additional flux constraints from inverse muon decay. SAND may also

make complementary measurements of $\nu + e$ elastic scattering, with $\sim 10\times$ lower statistics but better momentum and angle resolution compared to ND-LAr.

ND-X10: While it is possible to satisfy the requirement to monitor the neutrino beam by ensuring that ND-LAr+TMS makes regular on-axis measurements, and is able to quickly move on-axis in the event that a change is detected by the muon monitor system in order to characterize the impact on the neutrino spectrum, this capability can also be fulfilled by SAND. With a permanent on-axis detector, ND-LAr+TMS can take off-axis data for extended periods of time. This reduces the total amount of movement; instead of going back and forth between on-axis and a particular off-axis position, ND-LAr+TMS can perform an off-axis scan, including for an entire beam run. This builds in operational flexibility, while improving the on-axis beam monitoring by making it truly continuous. Compared to ND-LAr+TMS, SAND also has better resolution to the flux tail because it can measure higher energy muons by curvature.

4 Beam Monitoring

4.1 Introduction

The intensity of the primary proton beam, combined with other extreme conditions, make conventional neutrino beamlines extremely volatile environments. This is especially the case for the LBNF beam line given the unprecedented proton beam power that will be used to generate the neutrino beam. The stability and monitoring of the LBNF neutrino beam is essential to the physics goals associated with the ND, including the long baseline neutrino oscillation program, since they depends on understanding the neutrino flux, which may be impacted by potential variations.

The associated monitoring of the beamline has several purposes (*e.g.* beamline alignment, hardware protection, component performance, mitigation of issues arising from radiation, etc.); here we will focus on the physics impacts arising from variations in the neutrino flux resulting from changes and variations in the LBNF neutrino beam line. In this context, the primary goal of the beamline monitoring is to minimize beam operations and data-taking that may be compromised by unknown operational variations that impact the neutrino flux and thus result in systematic errors. A secondary goal is to provide feedback on the origin and nature of such variations when detected by any of the beamline instrumentation.

4.2 LBNF Beamline and Targeting Instrumentation

The LBNF beamline includes many monitoring devices divided between those that monitor the primary proton beam line incident on the target (Primary Beamline Monitors) which include

- **Beam position monitors (BPMs):** 24 dual-plane that measure the proton beam trajectory on its path to the target.
- **Beam loss monitors (BLMs):** 30 ion chambers that detect beam losses at the quadrupole magnets.
- **Beam intensity monitors (BIMs):** toroidal current transformers which measure the net proton flux.
- **Beam profile monitors:** secondary emission monitors that measure the transverse profile of the beam along two axes.

and those that monitor the status and activity in subsequent elements (Neutrino Beamline Monitors (NBI)):

- **Cross-hair monitors:** beryllium, aluminum, and graphite strips traversing the apertures of the baffle and horns from which particle production is detected when the proton beam passes through them, providing *in situ* alignment information.
- **Hadron alignment detector system (HADeS):** a retractable 13×13 grid of ionization detectors situated at the end of the decay pipe, just upstream of the hadron absorber
- **Target position thermometer (TPT):** beryllium rods positioned in front of the target that determine the alignment of the target face relative to the primary proton beam via the temperature profile measured on the rods.
- **Horn Location System (HLS):** a system of hydrostatic level sensors and linear variable differential transformers which measure the vertical position and tilt of the horns.
- **Muon Monitor System (MuMS):** three planes of 9×9 pixelated ionization detectors situated in alcoves with upstream shielding which stop muons with energies less than 5, 11 and 14 GeV, that measure the flux and transverse profile of muons with greater energy in each alcove.

Additional elements, such as horn current transformers, monitor other parameters that are critical to the behavior and performance of the beam line. We will collectively refer to all the instrumentation within the beamline as Beamline and Targeting Instrumentation (BTI). Some of the BTI elements, most notably the BPMs, BLMs, TPT, HLS, and MuMS provide continuous and near (if not actual) pulse-by-pulse feedback, while others (Beam Profile Monitors, BLMs, cross hairs, HADeS) require dedicated beam line configurations and deployment, most notably as part of the Beam-Based Alignment (BBA) procedure. For detecting and diagnosing many potential variations (*e.g.* a change in the primary proton beam, alignment of the components, *etc.*), the beamline instrumentation is the primary and definitive means.

4.3 Neutrino-based Beam Monitoring:

4.3.1 Motivation

A neutrino-based beam monitoring system (NuMS) in which neutrino interactions are used directly to assess the status and stability of the beam, serves several purposes in relation to BTI:

- Diagnostics when signals from BTI are subtle or ambiguous, especially in distinguishing between variations with different implications for the resulting neutrino flux.
- Directly informing whether potential beamline variations have impacted the neutrino flux
- Assess long-term trends associated with component degradation or drift
- Complement information provided by BTI that require dedicated deployment/running that is not continuously available through the run.

As an example of the first and third, a change in the primary proton beam profile and a horizontal drift in the angle of a horn would not have immediate BTI diagnosis apart from a change in the muon flux measured by MuMS. The former, however, typically has a small $< 1\%$ impact on the neutrino flux, while a horn angle shift could induce a sizable, $\sim$ few percent energy-dependent change. While additional elements of BTI, through the BBA procedure, would definitively diagnose the situation, this requires dedicated beam configurations and running that result in significant

down time. Thus, NuMS could not only confirm an indication from the MuMS, but provide further diagnostics that would inform subsequent steps in the beam operation. As an example of the second, a target degradation or other gradual deterioration of the beam line components, do not have dedicated BTI monitoring, but may induce significant changes of the neutrino flux over an extended period of months or more. While these may have signals in MuMS or other elements of BTI, regular monitoring directly of the neutrino events is needed to detect long term variations, particularly induced by unexpected component deterioration, to guide any intervention in the beam line operations.

4.3.2 On- vs. Off-axis

Variations are usually most readily detected through neutrino interactions on the beam axis due to a combination of the spectrum, where the neutrino energies most directly reflect the kinematics of the pion decays, and its intensity, where due to the maximum flux, statistics are most rapidly accumulated. In contrast, the spectrum of neutrinos observed off-axis increasingly break this correlation with the pion kinematics and have lower overall intensity. Hence, on-axis NuMS measurements are considered references that are conducted regularly to assess long-term trends, though in some cases, targeted off-axis detection may provide further diagnostic information. This is relevant given that ND-LAr+TMS will spend considerable time off-axis as part of its measurement program. This motivates a requirement to have periodic monitoring measurements on-axis where the diagnostic strength is usually maximized in the case that an dedicated on-axis NuMS is absent.

4.3.3 NuMS observables

Given that $\nu_\mu/\bar{\nu}_\mu$ are the dominant component of the flux in FHC/RHC mode, NuMS will be based on observing their interactions and measuring the following observables:

- **Overall Rate:** observed rate of neutrino interactions per proton-on-target, correlating to the overall neutrino flux produced by the beam line.
- **Transverse distribution:** vertex distribution of the neutrino interactions relative to the beam axis, sensitive to the alignment of beam line components internally and with respect to the near and far detectors.
- **Energy Spectrum:** distribution of a quantity correlating with the incident neutrino energy of the observed neutrino interactions, sensitive to the spectrum of the neutrino flux.

The periodicity and time frames for on-axis NuMS are motivated by minimizing potentially compromised beam running resulting from uncertainty in beam line conditions and sufficient granularity in timing to assess trends and provide feedback when the beamline operational parameters are changed. In formulating requirements, the required capabilities of NuMS should be separated from detailed run plans which would be formulated and optimized based on the conditions at any given time. For example, in the absence of a dedicated on-axis NuMS, it is anticipated that ND-LAr+TMS will remain on-axis to perform the NuMS role during potentially volatile periods such as beam start up periods.

4.4 NuMS requirements:

. The NuMS requirements follow from the following:

- The NuMS will operate on-axis at least once per month

- The NuMS will collect sufficient statistics for these measurements, resulting in a minimum fiducial mass
- The NuMS can move back on-axis promptly for beam monitoring purposes
- The NuMS will identify individual $\nu_\mu/\bar{\nu}_\mu$ CC interactions
- The NuMS will reconstruct the observables mentioned in Section 4.3.3, namely energy spectrum, and the vertex distribution in $\nu_\mu/\bar{\nu}_\mu$ CC events. An overall rate measurement follows from the total number of observed events.

In many cases, the resulting requirements overlap with or are supported by other capability requirements for ND-LAr+TMS and the PRISM movement system.

5 Capability requirements

The detector capabilities are motivated primarily by the physics measurement requirements given in Table 3, and in some cases by the desire to enable measurement goals given in Tables 4 and 5. This section is subdivided into four categories. Section 5.1 describes the capability requirements for the ND-LAr+TMS measurement system. Section 5.2 gives the requirements specific to the movement of that system. Section 5.3 describes the required capabilities of SAND, some of which are motivated by goals that leverage the KLOE magnet and calorimeter system. Section 5.4 describes the capabilities required for neutrino beam monitoring, which must be fulfilled by either ND-LAr+TMS or SAND.

5.1 ND-LAr+TMS capabilities

The capability requirements of ND-LAr with TMS as its muon spectrometer are given in four tables. The requirements in Table 6 derive from ND-M0, to match the FD detector performance with a LArTPC. Table 7 contains the requirements deriving from ND-M1, to measure neutrino-electron elastic scattering as a flux constraint. Table 8 gives the requirements from ND-M7, to separate activity from individual neutrino interactions. Finally, Table 9 defines the requirements on the TMS performance, which derive from ND-M0, ND-2, and ND-M7. An additional set of requirements described in Section 5.4 applies to ND-LAr+TMS only if it is required to serve as the beam monitor.

ND-C1.1: ND-LAr (with TMS as a muon spectrometer) satisfies requirement ND-M0, that neutrino interactions be reconstructed in the same way at the ND and FD. ND-C1.1.1 and ND-C1.1.2 pertain to electron and muon reconstruction for charged-current events. The requirement is to match or exceed the momentum resolution of the FD, and to avoid significant gaps in phase space coverage. Because of its large volume, the FD will contain the lepton in most interactions. In the ND, this is not necessary; rather, what is important is that the full phase space of interaction kinematics is measured and constrained by the ND. This means that efficiency corrections will be required, which will depend primarily on the geometry of the detector, and where the neutrino interaction occurs. ND-1.1.3 is to match the hadron reconstruction. Again, hadrons are almost always contained in the FD, and for the ND the full phase space must be measured. These requirements set the minimum dimensions of ND-LAr.

ND-C1.2: Measurements of charged-current interactions inherently constrain a convoluted product of flux and cross section. Because of oscillations, the flux and cross section uncertainties impact the near and far detector spectra differently. To break this degeneracy, it is necessary to measure

Label	Name	Requirement	Rationale	Ref. Req.
ND-C1.1	Match FD performance	ND-LArTPC shall classify and measure outgoing particles with performance matching the FD	To predict the FD spectrum as a function of rec. ν energy, ND-LAr must measure CC events similarly	ND-M0
ND-C1.1.1	ν_e CC e phase space	ND-LAr shall identify e from ν_e CC events covering $> 99\%$ of the interaction phase space of events selected at the FD	To constrain the predicted ν_e CC prediction at FD, ND-LAr needs to observe outgoing e over the same phase space.	ND-M0, ND-M3
ND-C1.1.2	ν_e CC e energy	ND-LAr shall measure e energy in ν_e CC events with $< 8\%$ resolution	To constrain . . . , ND-LAr needs to reconstruct e energy with comparable resolution or better.	ND-M0, ND-M3
ND-C1.1.3	ν_e CC e identification	ND-LAr shall separate e from ν_e CC events from γ	To constrain . . . , ND-LAr must reject backgrounds to e .	ND-M0, ND-M3
ND-C1.1.4	ν_μ CC μ phase space	ND-LAr+TMS shall identify μ from ν_μ CC events covering $> 99\%$ of the interaction phase space observed at the FD	To constrain the predicted ν_μ CC prediction at FD, ND-LAr needs to observe outgoing μ over the same phase space.	ND-M0
ND-C1.1.5	ν_μ CC μ energy	ND-LAr+TMS shall measure μ energy in ν_μ CC events with $< 5\%$ resolution in the oscillation peak region	To constrain . . . , ND-LAr needs to reconstruct μ energy with comparable or better resolution to the FD.	ND-M0
ND-C1.1.6	Hadron phase space	ND-LAr shall provide full containment of hadrons for 99% of the interaction phase space of events selected at FD.	As FD measures hadrons by calorimetry in LAr, to constrain the predicted $\nu_{e/\mu}$ CC prediction at FD, ND-LAr needs to observe the hadronic recoil in these events over the same phase space.	ND-M0
ND-1.1.7	Hadron response	For contained hadrons, ND-LAr shall match or exceed the FD tracking efficiency, and differences in calorimetric response shall be understood to $< 5\%$	To predict the reconstructed spectrum at the FD, ND-LAr must measure the hadronic component at least as well, with equal or better thresholds	ND-M0

Table 6: ND-LAr+TMS Capability Requirements for fulfilling ND-M0 matching FD performance

Label	Name	Requirement	Rationale	Ref. Req.
ND-C1.2	Measure flux normalization	ND shall measure the neutrino flux normalization with $< 2\%$ uncertainty	Justification	ND-M1
ND-C1.2.1	$\nu - e$ fiducial mass	ND shall have a fiducial mass capable of identifying at least 2500 $\nu - e$ scattering events/year	Sufficient statistics to reach 2(1)% statistical uncertainty in 1(4) years is needed.	ND-M1
ND-C1.2.2	$\nu - e$ e identification	ND shall separate the e in $\nu - e$ scattering events from γ	EM backgrounds from π^0 decay and other sources of photons must be eliminated to isolate $\nu - e$ scattering events (this requirement is redundant with ND-C1.1.3)	ND-M1
ND-C1.2.3	$\nu - e$ $E\theta^2$ measurement	ND shall measure $E\theta^2$ of the outgoing e in $\nu - e$ with sufficient resolution to separate it from ν_e CC events	To isolate a pure sample of $\nu - e$, it is necessary to use the forward kinematic signature of this interaction, encapsulated in $E\theta^2$ for the e .	ND-M1
ND-C1.2.4	$\nu - e$ vertex activity	ND shall identify energy depositions from protons > 20 MeV emitted at the vertex	To isolate a pure sample of $\nu - e$, it is necessary to use its topological signature of a single unaccompanied e , whereas ν_e CC backgrounds will have hadron emission or activity from nuclear breakup.	ND-M1

Table 7: ND-LAr Capability Requirements to fulfill ND-M1 (constrain the flux with $\nu - e$ scattering).

Label	Name	Requirement	Rationale	Ref. Req.
ND-C1.3	Identify individual ν interactions	ND-LAr+TMS shall separate activity from individual neutrino interactions	The FD will observe single beam neutrino events given the negligible pileup. The ND must disambiguate and correctly associate activity from the many interactions that occur within a single beam spill.	ND-M0, ND-M7
ND-C1.3.1	ND-LAr 3D charge readout	ND-LAr shall employ 3D charge readout using pixels	Due to pile up, 3D readout is needed to spatially separate activity that would otherwise overlap in 2D projective readouts (wires, strips)	ND-M7
ND-C1.3.2	ND-LAr 3D charge voxelization	ND-LAr charge readout shall voxelize charge activity to $> 5\text{mm}$ in each dimension.	In order to match FD spatial resolution of 5 mm, charge voxelization, and hence the pixel pitch and time resolution should allow comparable or better resolution.	ND-M7
ND-C1.3.3	ND-LAr light readout	ND-LAr shall time-stamp activity with optical signals with resolution comparable to or exceeding the bucket spacing of 18.83 ns	In order to unambiguously associate energy depositions detached from the vertex (<i>e.g.</i> neutrons) in a neutrino event, timing signals with resolution better than the bunch spacing are needed.	ND-M7
ND-C1.3.4	ND-LAr modularity	ND-LAr shall have segmented optical volumes that have < 10 separate interactions in a spill.	Due to the limited ability to localize signals, their multiplicity within a single optical detection volume must be limited in order to avoid an intractable many-to-many matching problem in associated them with charge signals.	ND-M7

Table 8: ND-LAr requirements to fulfil ND-M7 (pileup mitigation/single neutrino reconstruction requirements).

Label	Name	Requirement	Rationale	Ref. Req.
ND-C1.4	TMS μ re-con.	TMS shall detect/reconstruct μ produced by ν_μ CC in the ND-LAr fiducial volume that exit ND-LAr to support ND-C1.1.4, ND-C1.1.5	μ reconstruction is essential to reconstruct ν_μ CC events and constrain predictions in FD.	ND-C1.1.4, ND-C1.1.5
ND-C1.4.1	TMS μ phase space	TMS shall detect and analyze μ originating in the ND-LAr fiducial volume with $T_\mu \gtrsim 500$ MeV and $\theta_\mu \gtrsim 30^\circ$	In order to satisfy ND-C1.2.2, TMS must reconstruct muons that are produced by ν_μ CC interactions in the ND-LAr fiducial volume that exit ND-LAr, without a gap in kinematic coverage.	ND-C1.1.4
ND-C1.4.2	TMS μ resolution	The measurement of T_μ by μ range shall match or exceed the FD ($\sim 5\%$)	Measurement of the μ energy is an essential component of the reconstruction of ν_μ CC events	ND-C1.1.5
ND-C1.4.3	TMS timing resolution and synchronization	TMS shall have hit timing resolution and synchronization with ND-LAr comparable to or exceeding the bucket spacing of 18.83 ns	TMS must have timing resolution sufficient to separate activity from different neutrino interactions and to match with ν_μ CC interactions observed in ND-LAr (ND-C1.3.2)	ND-M0, ND-M7
ND-C1.4.4	TMS spatial resolution	TMS shall have sufficient 2D spatial resolution to match ND-LAr rear-exiting μ tracks that enter the front face of TMS.	To reconstruct μ that exit ND-LAr into TMS, signatures from a single μ in ND-LAr and TMS must be matched	ND-M0, ND-M7
ND-C1.4.4	TMS μ stopping power	TMS shall have sufficient stopping power to range out 5 GeV muons.	In order to reconstruct μ energy by range, it must be ranged out. A 5 GeV maximum allows of ν_μ CC with energy up to ~ 5 GeV, the high end of the oscillation region of interest, to be covered.	ND-M0
ND-C1.5	TMS μ sign selection	TMS shall separate μ^- and μ^+	Sign selection allows $\nu_\mu/\bar{\nu}_\mu$ CC interactions to be separated, thus allowing constraints on “wrong-sign” cross-contamination.	ND-M0, ND-M2

Table 9: TMS capability requirements to fulfil ND-M0, ND-M2, ND-M7

neutrino-electron elastic scattering, a purely electroweak process with a calculable cross section at tree level. The cross section is tiny, which sets the required fiducial mass of the detector at 25 tons, in order to obtain a sample such that the total statistical uncertainty is $< 2\%$. The electron identification requirement in ND-C1.2.2 is redundant with ND-C1.1.1. The energy and angle resolution for electrons is driven by this sample. The angle reconstruction is especially important, because the kinematic limit for $\nu + e$ is $E\theta^2 < 2m_e$, where m_e is the mass of the electron. It is also important to be able to cut hard on evidence of nuclear breakup in the form of nuclear fragments near the vertex. ND-C1.2.4 requires that a proton with 20 MeV kinetic energy be detectable; this does not mean that it must be tracked, but that its energy around the vertex can be observed so that it can be rejected.

ND-C1.3: From ND-M0, ND-LAr must match the performance of the FD, which requires isolating single neutrino interactions. From ND-R5 and ND-M7, ND-LAr must be able to operate in a high rate environment. This imposes requirements on ND-LAr that go beyond the FD, which does not have a high rate. Charge drifts for several 100s μs in ND-LAr, which is far longer than the duration of the spill; this means that ND-LAr essentially measures all charge deposits at once, and cannot separate them using timing.

The 3D charge readout requirement ND-C1.3.1 ensures that tracks and showers do not overlap. Even at 7.5×10^{13} protons per spill, particles in ND-LAr do not overlap in three dimensions; however, in any 2D projection there is a very high degree of overlapping activity. A second challenge is associating detached activity to its neutrino interaction. For example, a neutron can travel ~ 1 m in ND-LAr and then initiates a visible hadronic shower. That activity might be spatially proximate to several different neutrino vertices. The association can be performed using the timing information in the light detector. However, in a monolithic detector, the number of charge and timing signals would be ~ 50 , and the combinatorics of charge-light matching is impossible. ND-C1.3.3 and ND-C1.3.4 ensure that this association can be done by requiring a modular light detector, such that the number of interactions in a single light detector is < 10 .

ND-C1.4: TMS must supplement ND-LAr with muon reconstruction, such that the overall event reconstruction of the ND-LAr+TMS measurement system matches or exceeds the performance of the FD. As in ND-C1.1, the required acceptance is to avoid gaps in kinematic phase space; this means the TMS dimensions must be large enough to provide full angular coverage. For fiducial ND-LAr events, the minimum energy to reach TMS is about 500 MeV, and muons of this energy are preferentially forward. Muons at very wide angles are reconstructed by containment in ND-LAr. The overall momentum resolution must match the $\sim 5\%$ of the FD. For muons that barely reach the TMS, the resolution to the TMS portion of the track may be worse than 5%, because most of the energy is deposited in ND-LAr. The timing resolution of TMS must be sufficient to separate activity from different neutrino interactions using timing. The total length of TMS in the beam direction must be sufficient to stop a 5 GeV muon, because this is the upper end of the oscillation energy range of interest.

ND-C1.5: The TMS must have sign selection capability so that ND-LAr+TMS can measure wrong-sign interactions. This is important because the wrong-sign ν_μ at the ND oscillate to ν_e at the FD differently, and measuring this asymmetry is central to DUNE's oscillation physics program.

5.2 DUNE-PRISM capabilities

The requirements related to the movement of ND-LAr+TMS as part of DUNE-PRISM are given separately in Table 10, because they define the movement system rather than the detectors themselves. The full program of off-axis measurements requires the core detector functionality described in Table 6 in addition to the movement system.

Label	Name	Requirement	Rationale	Ref. Req.
ND-C2.1	Travel distance	PRISM shall move ND-LAr+TMS to at least 28 meters off-axis	PRISM must allow off-axis data taking to at least 30.5 me off-axis to cover the energy range of the second oscillation peak. The fiducial volume of ND-LAr is 3 m wide so its midpoint must be able to move up to 28 m.	ND-M4
ND-C2.2	Uniform detector performance	ND-LAr and TMS detector performance must be consistent following PRISM movements	Consistency in detector performance is needed to combine/compare data taken at different positions without additional corrections.	ND-M4
ND-C2.3	Detector placement accuracy	PRISM shall place ND-LAr+TMS within 10 cm of its desired position.	PRISM must place the detector elements accurately enough such that the neutrino spectrum matches the targeted spectrum at the desired off-axis position.	ND-M4
ND-C2.4	Detector placement measurement	PRISM shall measure the ND-LAr+TMS off-axis position to within 1 cm	The placement of ND-LAr+TMS must be accurately known to predict the neutrino flux at a given off-axis position	ND-M4
ND-C2.5	Movement downtime	The detector downtime between PRISM movements period shall be < 8 hours (including ramp down/up)	A complete movement cycle should be completed within a standard work shift to minimize complexity and downtime	ND-M4, ND-M6
ND-C2.6	Measurement cycle	The PRISM system shall allow at least 10 movements each operational year	A complete set of off-axis measurements, requiring $\gtrsim 10$ movements given the 3 m width of ND-LAr fiducial volume, is needed to avoid combining data across years, which may have considerable variations in the operating conditions.	ND-M4

Table 10: PRISM capability requirements

ND-C2.1: The maximum travel distance defines the minimum energy that can be accessed with DUNE-PRISM, because the flux peak decreases with increasing off-axis angle. To cover down to 0.5 GeV, which spans the first and second oscillation maxima, requires taking data out to 30.5 m. This can be achieved by moving the center of the detector 28 m, and then subdividing the volume into slices of different off-axis position within each movement stop.

ND-C2.2: The reconstruction performance must be identical at each off-axis position, with differences at the percent level or smaller. This ensures that differences between data taken at different positions is truly due to flux differences, and not some other performance effect.

ND-C2.3-5: These pertain to the mechanics of moving the detector. The positioning of the detector at each stop must be within 10 cm of the desired position. Once the detector is in place, a survey of its position is taken, with a required precision of 1 cm on the off axis position. The movement rate of 8 hours total is driven by the desire to be able to move the detector without a planned downtime of the beam, and without missing significant beam data. To control for possible changes in the beam over long periods of time, it is necessary to be able to take a full suite of on-axis and off-axis measurements within one year.

5.3 SAND capabilities

The capability requirements for the SAND physics goals are given in Table 11. The capabilities required for SAND to fulfill the beam monitoring requirement are discussed in Section 4.4. SAND contributes to the primary neutrino oscillation physics measurement program in two distinct ways. First, as a permanent, magnetized on-axis detector, it fulfills the neutrino beam spectrum monitoring requirement ND-R4, such that the movable ND-LAr+TMS system does not need to make regular on-axis stops to perform beam monitoring. Second, it measures neutrino interactions in ways that complement the required ND-LAr+TMS measurement program. SAND cannot ever fulfill ND-M0, the requirement to measure neutrino-argon interactions in a LArTPC to predict the FD spectra. SAND can measure neutrino interactions on both argon and “solid hydrogen” targets, with a detector technique that is fundamentally different from ND-LAr, providing complementary information. Thus, while ND-LAr+TMS and the PRISM movement capability should be considered the fundamentally required components of the Phase I ND, SAND enhances the long-baseline physics reach, and also extends the physics reach of DUNE more broadly with additional BSM measurements.

SAND is unlike ND-LAr+TMS in that a major component of it is repurposed from an existing detector. The design motivation of SAND is similarly different. While the design of ND-LAr+TMS is motivated by the minimum high-level physics requirements of DUNE, SAND is motivated by leveraging the KLOE magnet and calorimeter to enhance the overall DUNE physics program. Thus, SAND is not designed to a minimum requirement specification in the same way as ND-LAr+TMS. The magnet and calorimeter vastly exceed the minimum requirement for many physics measurements in SAND.

ND-C3.1-4: The high intensity of the LBNF beam, combined with the KLOE solenoid and performant calorimeter, make it possible to use a low density target in a neutrino beam. This adopts the detection strategy commonly used in collider experiments: precise particle tracking with a low-density material in a magnetic field, surrounded by calorimeters. Previous neutrino facilities have not had this luxury because the low density would require an enormous volume to achieve adequate event rate; however, in LBNF at 1.2 MW, a 5 ton tracker would observe > 5 million ν_μ CC events per year. Using thin passive targets is possible in such a detector due to the precise vertexing, enabling different nuclear targets including “solid hydrogen.” A search for heavy neutral leptons becomes very promising, because the signal (a neutral particle *decaying* in

Label	Name	Requirement	Rationale	Ref. Req.
ND-C3.1	Tracker density	The SAND inner tracker should have average density $\rho < 0.22\text{g/cm}^3$	Low density is essential to minimize multiple scattering and allow precision magnetic spectrometry	ND-X1, ND-X8
ND-C3.2	Tracker momentum resolution	The SAND inner tracker should have charged particle momentum resolution $< 5\%$ up to $p \sim 5\text{ GeV}/c$	Precise momentum measurements via curvature is needed to perform kinematic reconstruction.	ND-X1, ND-X3, ND-X8
ND-C3.3	Tracker radiation length	The SAND inner tracker should have < 1 radiation length from its center to the ECAL	Minimizing radiation lengths allows precise curvature measurements particularly for $e^\pm$	ND-X8
ND-C3.4	Tracker interaction length	The SAND inner tracker should have < 1 interaction length from its center to the ECAL	Minimizing interaction lengths minimizes secondary hadronic interactions in the tracker.	ND-X8
ND-C3.5	Tracker (hydro)carbon target	The SAND inner tracker should have both (hydrogen-rich) hydrocarbon and pure carbon target planes	Deploying a combination of pure carbon and hydrocarbon targets allows the isolation of interactions on free protons via subtraction.	ND-X5, ND-X6, ND-X7
ND-C3.6	Tracker nuclear targets	The SAND inner tracker should be configurable with other nuclear target planes	Deploying other nuclear target planes allows A -dependence of cross section properties to be studied	NC-X7
ND-C3.7	Tracker argon target	The SAND inner tracker volume should have an argon target	Deploying argon targets allows a direct comparison of $\nu - \text{Ar}$ interactions with $\nu - \text{H}$ measurements enabled in ND-C3/5.	ND-X6, ND-X7

Table 11: SAND Inner Tracker capability goals

the detector volume) scales with the volume while the background (a regular neutrino *scattering* in the detector volume) scales with mass. The ability to reject events with nuclear breakup on the carbon target is also excellent, so searches for processes that do not break the nucleus (HNL, tridents, $\nu + e$, etc.) are very pure. This type of inner tracker design is the optimal way to leverage the existing magnet and calorimeter, given the intensity of the beam.

ND-C3.5: Using CH_2 as the primary target, with a thin passive pure carbon target, makes a unique “solid hydrogen” measurement possible. Models of neutrino-nucleus interactions, like those used in the DUNE oscillation analysis, typically factorize scattering on free nucleons with the effect of the nuclear medium. One of the biggest complications in constructing these models is that most available data is for neutrino-nucleus, rather than neutrino-nucleon scattering. Data from hydrogen and deuterium bubble chamber experiments in the 1960s and 1970s are still widely used, despite having low statistics and poorly-understood beams. The solid hydrogen subtraction technique is a way to make high-statistics measurements of neutrino-proton scattering in a modern, well-characterized neutrino beam, without the safety hazards of bubble chambers.

ND-C3.6: The ability to measure different nuclei would facilitate comparisons of neutrino cross sections on different nuclear targets. A program of measuring “A-dependence” of neutrino cross sections was carried out by MINERvA. With greatly improved vertexing and low tracker density, SAND could substantially improve on these measurements. SAND also has the potential to make ratio measurements where the denominator is free protons.

ND-C3.7: Because argon is the target of the DUNE FD, it is especially interesting to include as a nuclear target in SAND. There are also several complementary measurements that SAND could make on argon, leveraging the magnetic field.

5.4 Neutrino beam monitoring capabilities

The neutrino-based beam monitoring requirements are described in Section 4.3. These requirements can be satisfied by either ND-LAr+TMS, or by SAND. The preferred solution, given in ND-X10, is continuous monitoring with a permanent on-axis detector, SAND. It is also possible to satisfy these requirements by taking regular on-axis data with ND-LAr+TMS.

Table 12 gives the capability requirements for SAND to act as the beam monitor. Beam monitoring is not a design driver for SAND; it is possible to fulfill these requirements with a very basic inner tracker, because the existing KLOE magnet and calorimeter greatly exceed the required magnetic field, timing, and energy resolution.

ND-C4.1: To fulfill the beam monitoring requirements, SAND must have sufficient fiducial mass to achieve the required statistics (ND-C4.1.1). The 5 ton inner tracker is also desired for statistics of the SAND physics program in Table 11. The momentum, vertex, and timing resolutions specified in ND-C4.1.2-4 are vastly exceeded by a low density tracker in the KLOE magnet and calorimeter, the design of which is driven by the goals in Table 11. This means that as long as SAND is operating, it satisfies the beam monitoring requirement ND-M6, and ND-LAr+TMS is therefore not required to make regular on-axis measurements.

If SAND is not present and operating in the beam, then the ND-LAr+TMS (including movement) capabilities required for ND-M6 are given in Table 13.

The requirement ND-C4.2.1 significantly restricts the flexibility in the run plan of ND-LAr, and is the main reason why it is preferable for SAND to satisfy the monitoring requirements. ND-C4.2.1 may actually drive the required movement speed, because it is required to be able to take far-off-axis data and then return to on-axis for monitoring purposes.

The other requirements are all less stringent than similar requirements for ND-LAr+TMS for neutrino oscillation physics. ND-C4.2.2 requires a 5 ton fiducial mass, but ND-C1.2.1 requires

Label	Name	Requirement	Rationale	Ref. Req.
ND-C4.1	Continuous on-axis beam monitoring	SAND should continuously perform on-axis neutrino beam monitoring as described in ND-M6	If SAND is performing beam monitoring continuously on-axis, ND-LAr+TMS do not need to make periodic on-axis beam monitoring measurements as required by ND-M6.	ND-M6, ND-X9
ND-C4.1.1	SAND NuMS statistics	The SAND inner tracker should have a fiducial mass of at least 5 tons	With this mass, the inner tracker can provide a precise measure of the ν_μ CC energy spectrum in one week	ND-M6, ND-X9
ND-C4.1.2	SAND NuMS μ mom. resolution	The SAND inner tracker should have μ momentum resolution $< 5\%$ at 1 GeV/ c and $< 10\%$ at 5 GeV/ c	The SAND μ momentum resolution should be sufficient to precisely measure its spectrum.	ND-M6, ND-X9
ND-C4.1.3	SAND NuMS vertex resolution	The SAND inner tracker should measure neutrino vertex locations to < 5 cm	The SAND inner tracker should localize neutrino interaction in order to measure the beam spectrum as a function of beam position, which elucidates some beam line variations.	ND-M6, ND-X9
ND-C4.1.4	SAND NuMS timing resolution	The SAND inner tracker should have < 10 ns timing resolution	SAND should have sufficient time resolution resolve activity from different beam buckets (18.83 ns)	ND-M6, ND-X9

Table 12: SAND Neutrino Beam Monitoring Capability requirements.

Label	Name	Requirement	Rationale	Ref. Req.
ND-C4.2.1	Periodic on-axis beam monitoring	ND-LAr+TMS shall be capable of performing on-axis neutrino beam monitoring at least once a month	Monthly checks of the on-axis beam conditions mitigate the risk of unusable FD data due to unexpected beam variations	ND-M6
ND-C4.2.2	Neutrino monitor statistics	ND-LAr+TMS shall have a fiducial mass of at least 5 tons	With this mass, the ND-LAr+TMS can provide a precise measure of the muon energy spectrum in one week	ND-M6
ND-C4.2.3	Neutrino monitor μ mom. resolution	ND-LAr+TMS shall have μ energy resolution $< 10\%$ for $E_\nu \sim 1 - 5 \text{ GeV}/c$	The μ momentum resolution needs to be sufficient to precisely measure its spectrum.	ND-M6
ND-C4.2.4	Neutrino monitoring vertex resolution	ND-LAr+TMS shall measure neutrino vertex locations to $< 20 \text{ cm}$	ND-LAr+TMS needs to determine the neutrino interaction vertex in order to measure the beam spectrum as a function of beam position, which elucidates some beam line variations.	ND-M6, ND-X9

Table 13: ND-LAr+TMS Neutrino Beam Monitoring Capability requirements.

~ 25 tons for neutrino-electron elastic scattering statistics. Similarly, ND-C4.2.3 is less stringent than ND-C1.1.5 on the muon momentum resolution. While vertex resolution is not explicitly a requirement for oscillation physics, matching the hadron reconstruction of the FD (ND-C1.1) requires $\sim 5 \text{ mm}$ charge readout, and achieving a vertex resolution of 20 cm with that spatial resolution is trivial, and has been vastly exceeded in previous experiments with similar pitch.